$\int_0^\infty e^{-ax^2}\,dx = \frac{1}{2}\left(\frac{\pi}{a}\right)^{1/2}$	$\int_0^\infty xe^{-ax^2}\,dx = \frac{1}{2a}$
$\int_0^\infty x^2e^{-ax^2}\,dx = \frac{1}{4a}\left(\frac{\pi}{a}\right)^{1/2}$	$\int_0^\infty x^3e^{-ax^2}\,dx = \frac{1}{2a^2}$
$\int_0^\infty x^{2n}e^{-ax^2}\,dx = \frac{1\cdot3\cdot5\cdot\ldots\cdot(2n-1)}{2^{n+1}a^n}\left(\frac{\pi}{a}\right)^{1/2}$	$\int_0^\infty x^{2n+1}e^{-ax^2}\,dx = \frac{n!}{2a^{n+1}}$
$\int_{-\infty}^\infty x^{2n}e^{-ax^2}\,dx = 2\int_0^\infty x^{2n}e^{-ax^2}$	$\int_{-\infty}^\infty x^{2n+1}e^{-ax^2}\,dx = 0$
$\int_0^\infty x^n e^{-ax}\,dx = \frac{n!}{a^{n+1}}$	

Table 1: Table of Useful Integrals